

Documentation of basics networking concept in practice

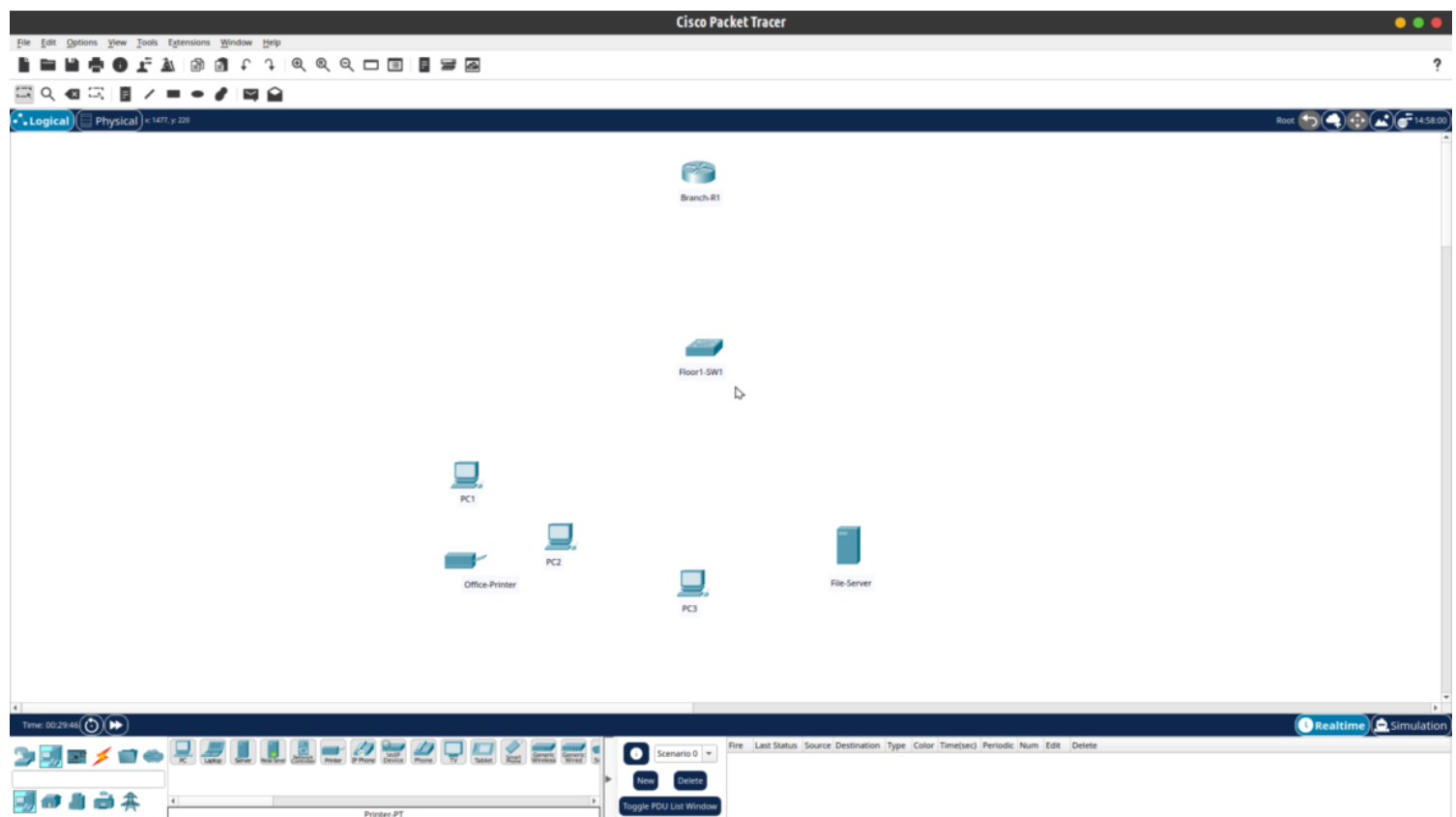
Skills to learned

- Device roles
- Cabling and interfaces
- Cisco CLI
- Packet analysis
- Basics switch and router interface configuration

Tasks:

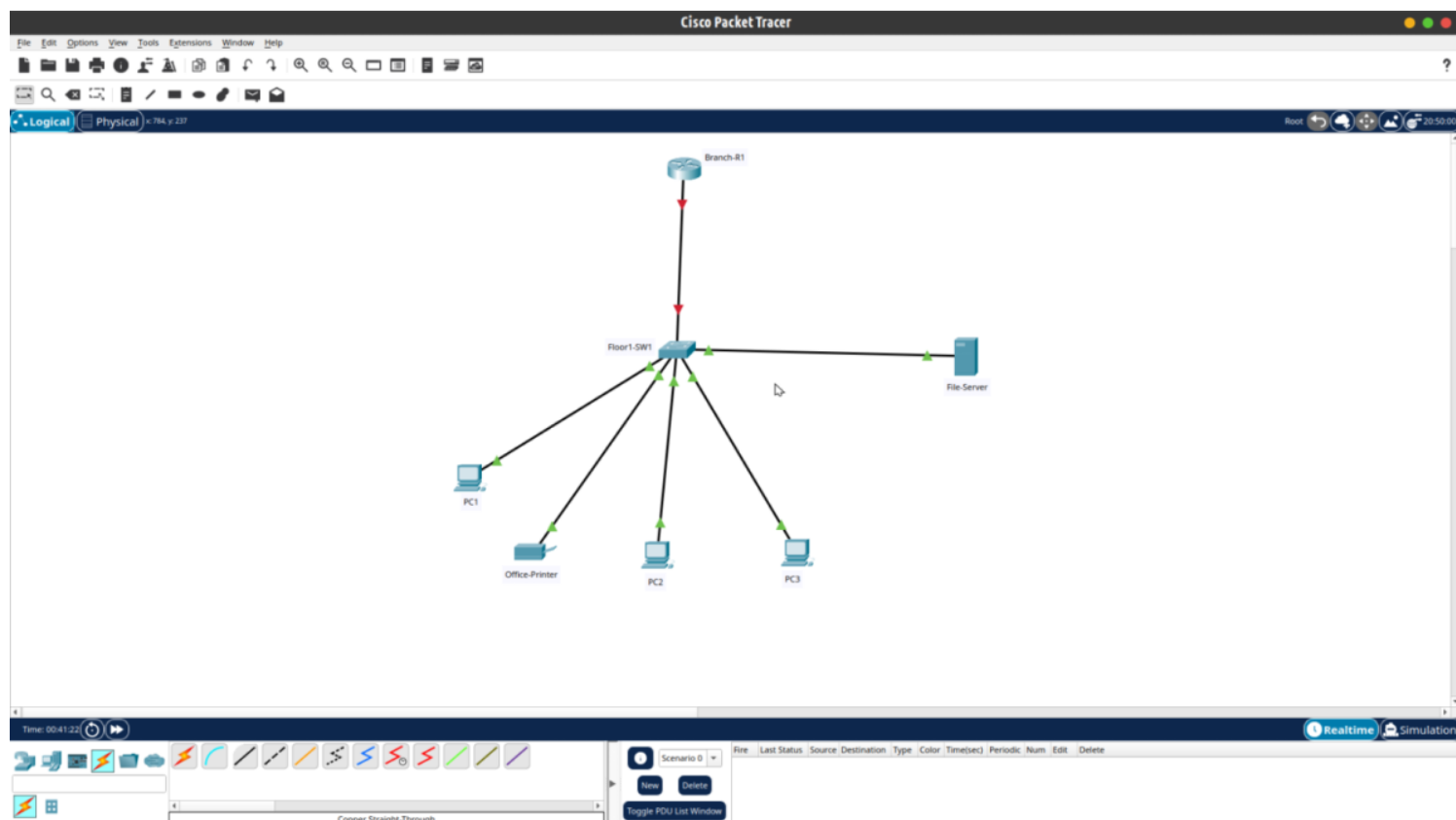
Step 1:

Name and configure device name on the system



Step 2:

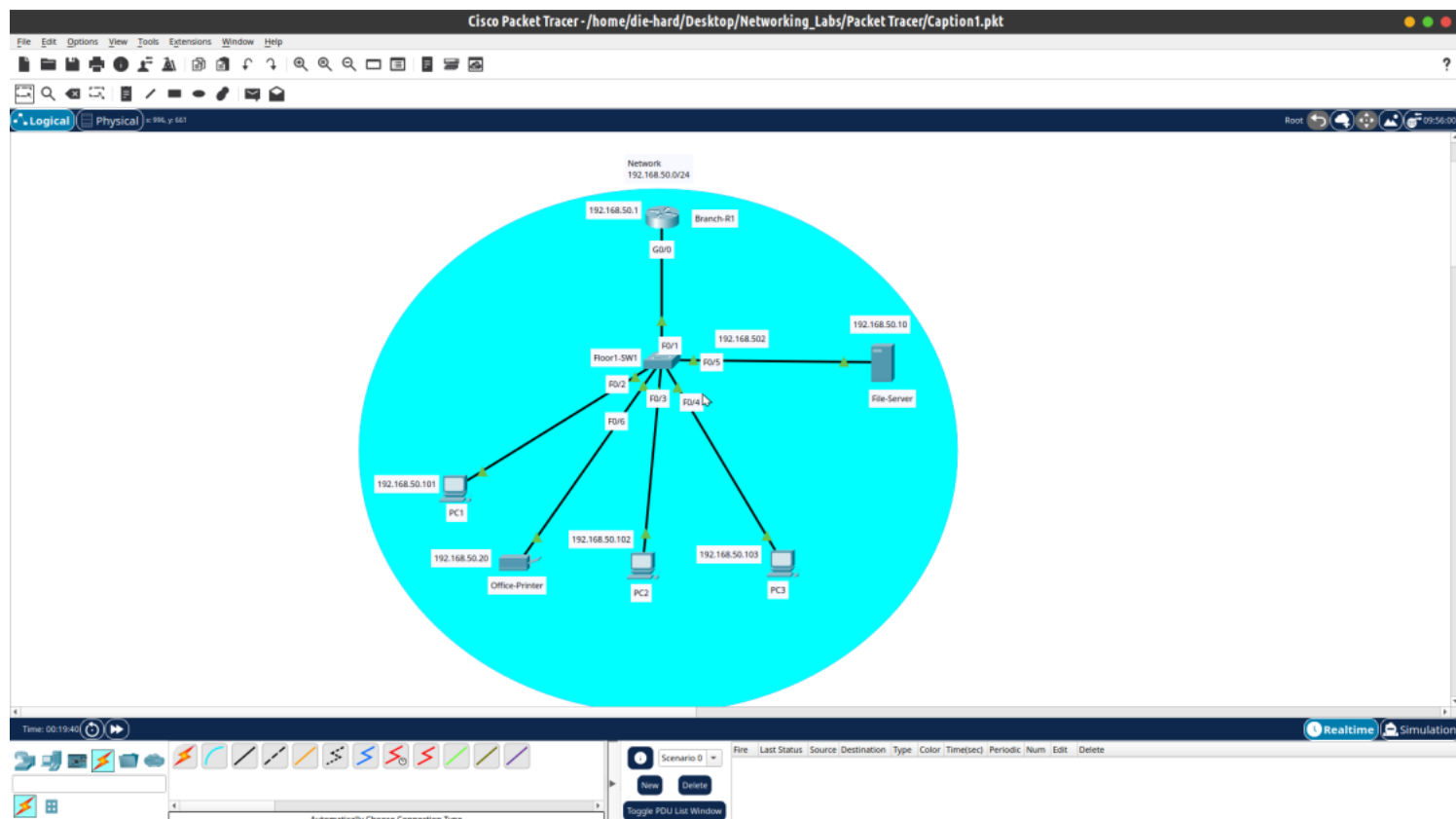
Choose the right cable to connect devices



Step3:

Configure the ip address in each end devices and network devices

S



Here are the IP addressing plan

Device	ip
Switch VLAN2	192.168.50.2
Server	192.168.50.10
Printer	192.168.50.20
PC1	192.168.50.101
PC2	192.168.50.102
PC3	192.168.50.103
Router	192.168.50.1

Device Naming and Location

Device	Hostname	Location
Router	Branch-R1	Server Room
Switch	Floor1-SW1	Floor1 Rack
Server	File-Server	Office
Printer	Office-Printer	Reception

Cable Mapping

From	TO
Router G0/0	Switch F0/1
PC1	Switch F0/2
PC2	Switch F0/3
PC3	Switch F0/4
Server	Switch F0/5
Printer	Switch F0/

Step 4:

Configure the router interface connected to the switch

Hostname, IP Address, Password Protected Router and Switch and enable the interface

Output from *show ip interface brief*

Branch-R1

Physical

Config

CLI

Attributes

IOS Command Line Interface

Branch-R1>
Branch-R1>
Branch-R1>
Branch-R1>
Branch-R1>
Branch-R1>
Branch-R1>
Branch-R1>
Branch-R1>
Branch-R1>
Branch-R1>en
Branch-R1>enable
Password:
Branch-R1#show ip interface brief
Interface IP-Address OK? Method Status
Protocol
GigabitEthernet0/0 192.168.50.1 YES manual up
up
GigabitEthernet0/1 unassigned YES unset
administratively down down
GigabitEthernet0/2 unassigned YES unset
administratively down down
Vlan1 unassigned YES unset
administratively down down
Branch-R1#
Branch-R1#
Branch-R1#
Branch-R1#
Branch-R1#

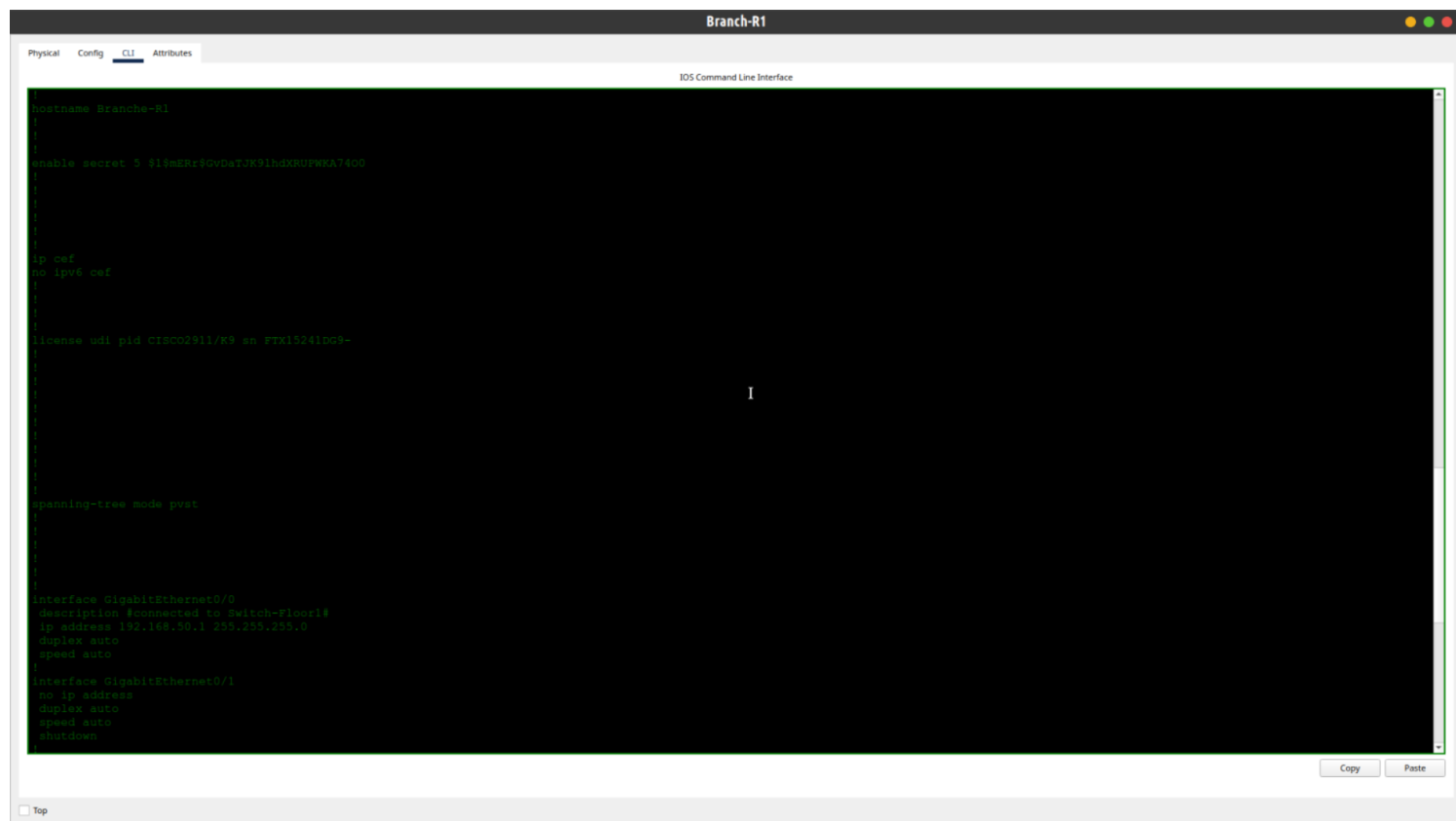
Copy

Paste

☐ Top

Output from *show running-config*

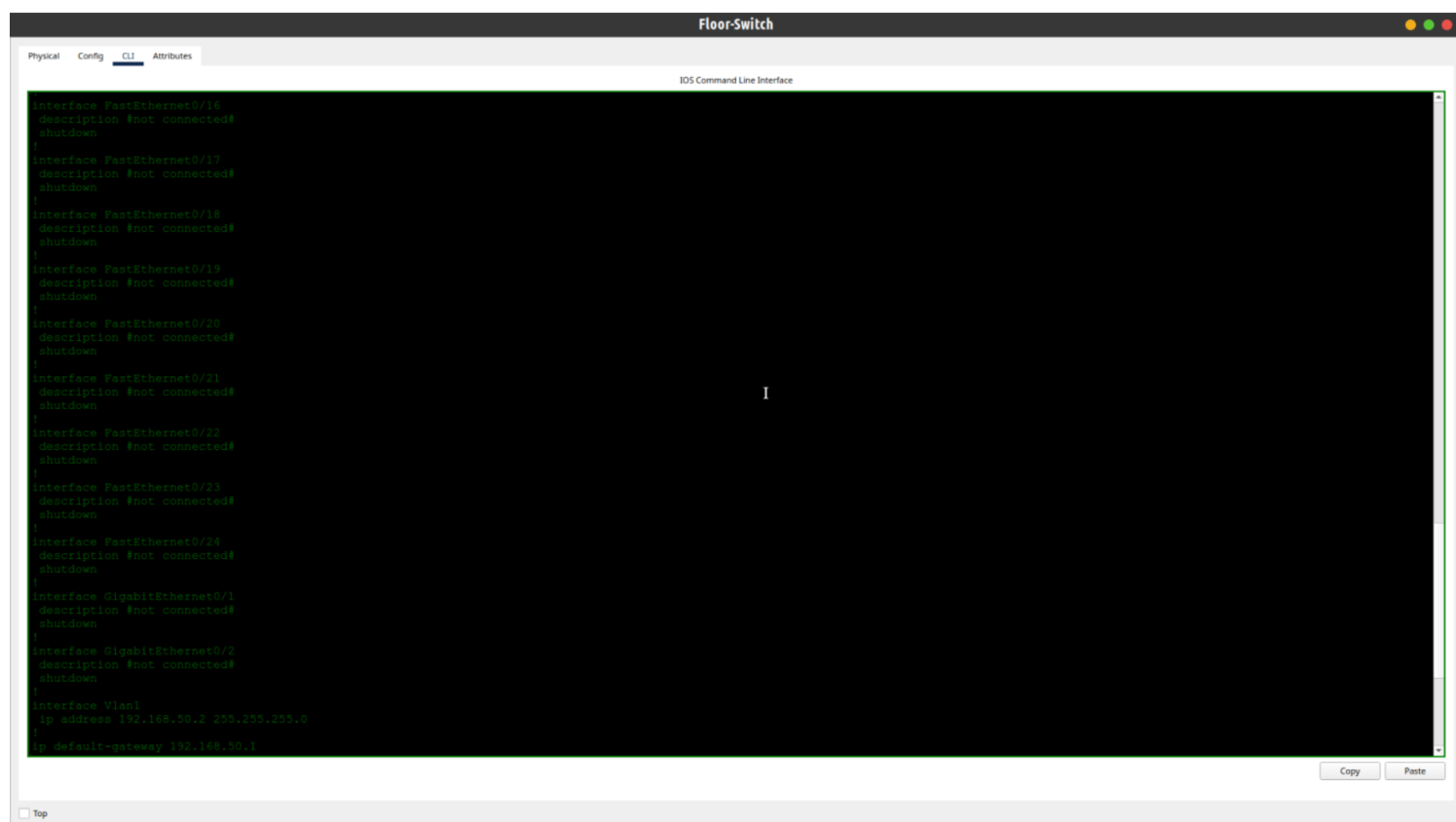
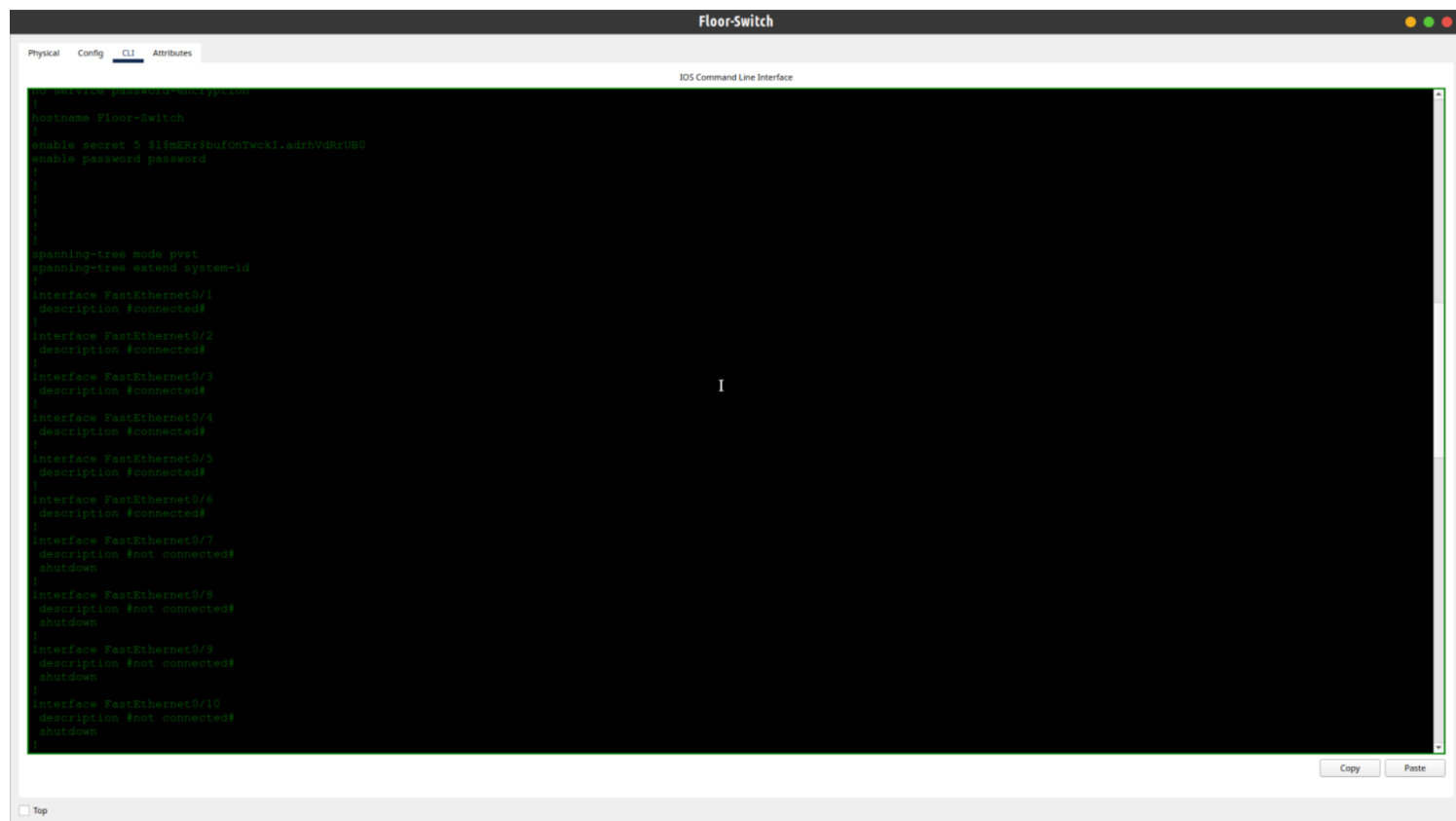
4/16



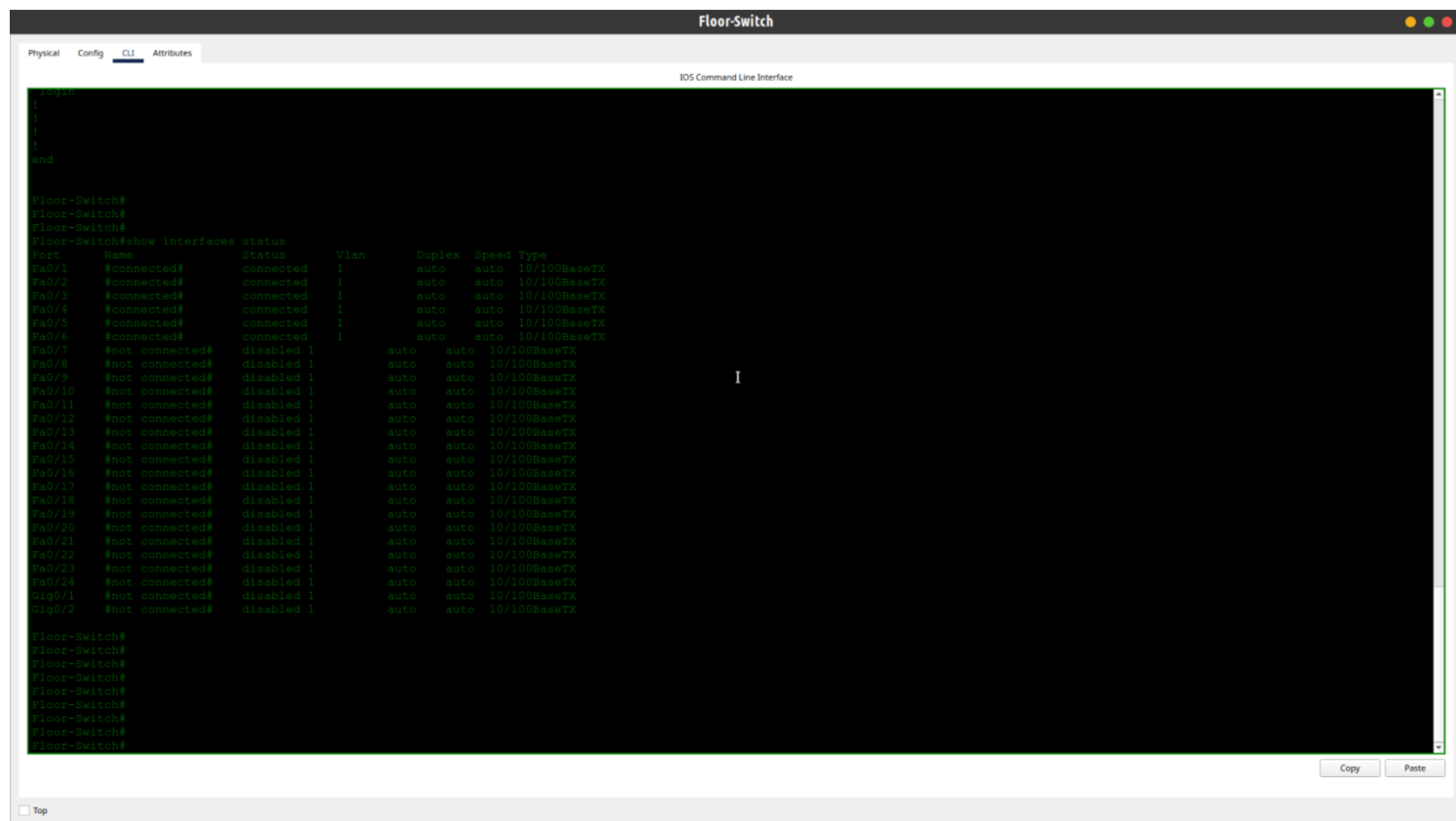
Step 5:

Configure Switch Hostname, IP Address on the vlan1 interface and default gateway in order to manage the switch and configure necessary configuration on each interface

Output from *show running-config*



Output from *show interfaces brief*

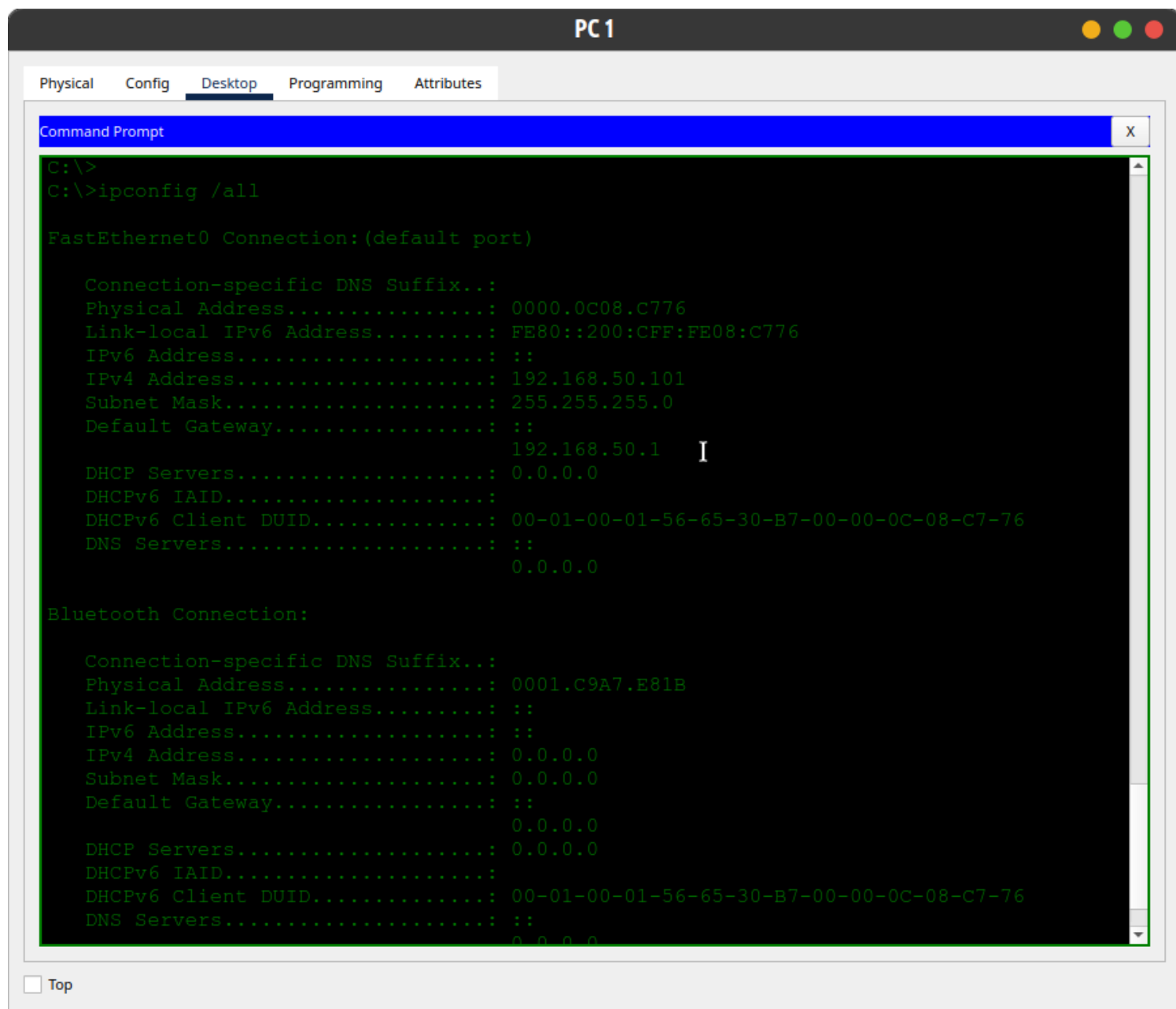


Step 6:

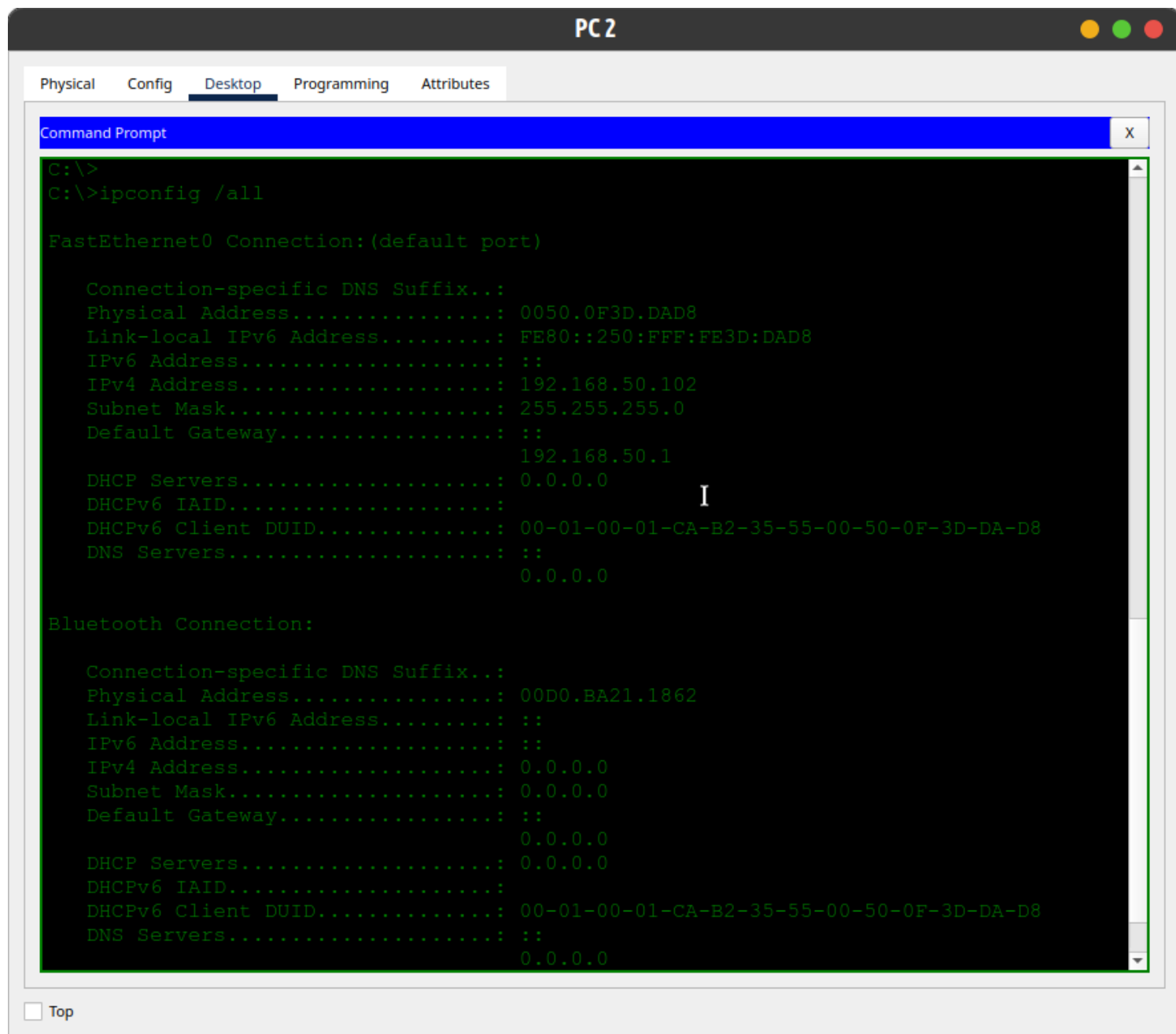
Configure the IP, Subnet Mask, Default Gateway on the End Device

Output from *ipconfig /all*

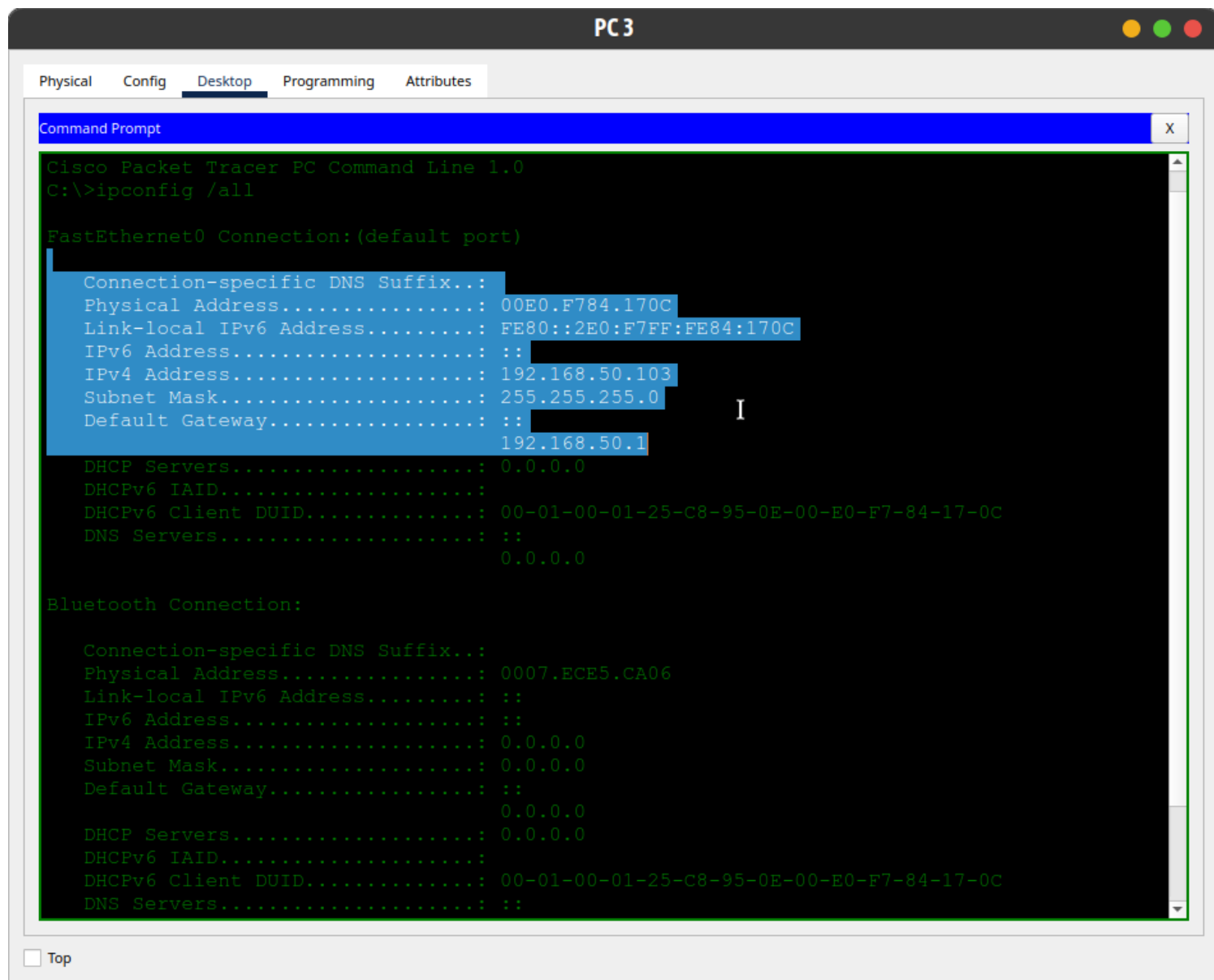
PC1



PC2



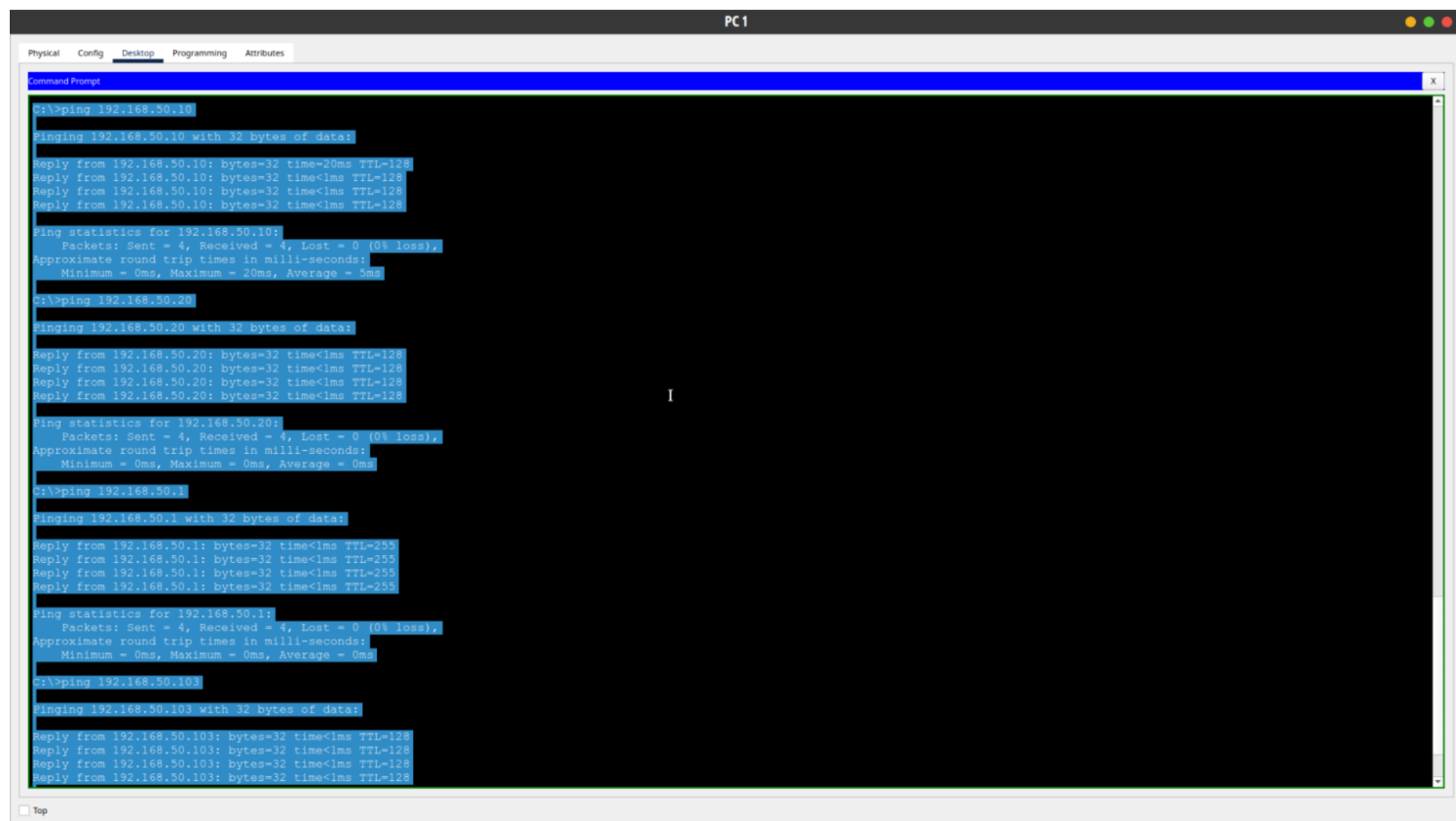
PC3



Step 7:

Test connectivity btw device to ensure communication can occur effectively

Output from *ping* command



Step 8:

Investigate the switch information including mac address table and how ports learned their mac address

Use the *show mac address-table dynamic*

Floor-Switch

PhysicalConfigCLIAttributes

IOS Command Line Interface

Floor-Switch#
Floor-Switch#show mac address-table
Mac Address Table

VlanMac AddressTypePorts

10030.f253.7b01DYNAMICFa0/1
Floor-Switch#show mac address-table
Mac Address Table

VlanMac AddressTypePorts

10000.0c08.c776DYNAMICFa0/2
10001.4340.5313DYNAMICFa0/5
10030.f253.7b01DYNAMICFa0/1
10050.0f3d.dad8DYNAMICFa0/3
100d0.d312.6165DYNAMICFa0/6
100e0.f784.170cDYNAMICFa0/4
Floor-Switch#
Floor-Switch#
Floor-Switch#
Floor-Switch#
Floor-Switch#
Floor-Switch#

CopyPaste

☐ Top

Step 9:

Monitor interface to check for active ports, unused, disable or disconnected ports

Use the *show interfaces status* cmd

Floor-Switch

PhysicalConfigCLIAttributes

IOS Command Line Interface

Floor-Switch#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1	#connected#	connected	1	auto	auto	10/100BaseTX
Fa0/2	#connected#	connected	1	auto	auto	10/100BaseTX
Fa0/3	#connected#	connected	1	auto	auto	10/100BaseTX
Fa0/4	#connected#	connected	1	auto	auto	10/100BaseTX
Fa0/5	#connected#	connected	1	auto	auto	10/100BaseTX
Fa0/6	#connected#	connected	1	auto	auto	10/100BaseTX
Fa0/7	#not connected#	disabled	1	auto	auto	10/100BaseTX
Fa0/8	#not connected#	disabled	1	auto	auto	10/100BaseTX
Fa0/9	#not connected#	disabled	1	auto	auto	10/100BaseTX
Fa0/10	#not connected#	disabled	1	auto	auto	10/100BaseTX
Fa0/11	#not connected#	disabled	1	auto	auto	10/100BaseTX
Fa0/12	#not connected#	disabled	1	auto	auto	10/100BaseTX
Fa0/13	#not connected#	disabled	1	auto	auto	10/100BaseTX
Fa0/14	#not connected#	disabled	1	auto	auto	10/100BaseTX
Fa0/15	#not connected#	disabled	1	auto	auto	10/100BaseTX
Fa0/16	#not connected#	disabled	1	auto	auto	10/100BaseTX
Fa0/17	#not connected#	disabled	1	auto	auto	10/100BaseTX
Fa0/18	#not connected#	disabled	1	auto	auto	10/100BaseTX
Fa0/19	#not connected#	disabled	1	auto	auto	10/100BaseTX
Fa0/20	#not connected#	disabled	1	auto	auto	10/100BaseTX
Fa0/21	#not connected#	disabled	1	auto	auto	10/100BaseTX
Fa0/22	#not connected#	disabled	1	auto	auto	10/100BaseTX
Fa0/23	#not connected#	disabled	1	auto	auto	10/100BaseTX
Fa0/24	#not connected#	disabled	1	auto	auto	10/100BaseTX
Gig0/1	#not connected#	disabled	1	auto	auto	10/100BaseTX
Gig0/2	#not connected#	disabled	1	auto	auto	10/100BaseTX

CopyPaste

☐ Top

Step10:

Simulating User Tickets

Ticket1

Checking what happen if an layer1 has an issue. *Interface shutdown*

Floor-Switch

Physical
Config
CLI
Attributes

IOS Command Line Interface

```

Floor-Switch(config-if)#exit
Floor-Switch(config)#do show interfaces status
Port      Name           Status      Vlan      Duplex  Speed  Type
Fa0/1     #connected#    connected   1         auto    auto   10/100BaseTX
Fa0/2     #connected#    connected   1         auto    auto   10/100BaseTX
Fa0/3     #connected#    disabled 1         auto    auto   10/100BaseTX
Fa0/4     #connected#    connected   1         auto    auto   10/100BaseTX
Fa0/5     #connected#    connected   1         auto    auto   10/100BaseTX
Fa0/6     #connected#    connected   1         auto    auto   10/100BaseTX
Fa0/7     #not connected# disabled 1         auto    auto   10/100BaseTX
Fa0/8     #not connected# disabled 1         auto    auto   10/100BaseTX
Fa0/9     #not connected# disabled 1         auto    auto   10/100BaseTX
Fa0/10    #not connected# disabled 1         auto    auto   10/100BaseTX
Fa0/11    #not connected# disabled 1         auto    auto   10/100BaseTX
Fa0/12    #not connected# disabled 1         auto    auto   10/100BaseTX
Fa0/13    #not connected# disabled 1         auto    auto   10/100BaseTX
Fa0/14    #not connected# disabled 1         auto    auto   10/100BaseTX
Fa0/15    #not connected# disabled 1         auto    auto   10/100BaseTX
Fa0/16    #not connected# disabled 1         auto    auto   10/100BaseTX
Fa0/17    #not connected# disabled 1         auto    auto   10/100BaseTX
Fa0/18    #not connected# disabled 1         auto    auto   10/100BaseTX
Fa0/19    #not connected# disabled 1         auto    auto   10/100BaseTX
Fa0/20    #not connected# disabled 1         auto    auto   10/100BaseTX
Fa0/21    #not connected# disabled 1         auto    auto   10/100BaseTX
Fa0/22    #not connected# disabled 1         auto    auto   10/100BaseTX
Fa0/23    #not connected# disabled 1         auto    auto   10/100BaseTX
Fa0/24    #not connected# disabled 1         auto    auto   10/100BaseTX
Gig0/1    #not connected# disabled 1         auto    auto   10/100BaseTX

```

Copy
Paste

☐ Top

Ping fails to reach f0/3 and *tracert* couldn't complete the first hop.
 Check interface status on switch and there was f0/3 interface disable
 Enable by running *no shutdown* on f0/3 interface

Ticket 2

Checking what happen if layer3 has an issue default gateway.

Defult gateway is a layer 3, local area network communication works pretty find but when it goes to beyond LAN there is problem

Ticket3

Checking what happen if subnet mask has an issue

Wrong subnet mask doesn't affect LAN communication

Step 11:

Packet analysis

Using Packet Tracer Stimulation Mode to check how packet flow using different protocol and how packet move from layer by layer doing encapsulation and de-encapsulation

When pc1 *ping* the server and ICMP Echo Request is send to the destination address where the switch receive and look into it MAC address table to see the intended device and if it MAC address is found the switch forward using unicast frame. From pc1 it encapsulate the packet and send to switch which de-encapsulate the packet and check for destination mac, encapsulate and foward it. When received by intended device, it decapsulate and check for destination IP and send the ICMP Echo Reply and communication establish. But if the MAC address of the intended device is not found on the switch MAC address table an ARP request is created where the switch forwarded a broadcast frame of ffff.ffff.ffff to all the device except where it received from and the asking for the IP address. The device that has the IP address replied and a unicast frame is sent to the device.

Note for FTP communication in packet tracer

- Configure ftp server by enabling FTP services
- Create account on the FTP server
- Connect you end device to the server
- Create file and upload on the server
- Retrieve the file from other end device

FTP at Work

When user want to retrieved file from the file server on their client machine a TCP connection is created, if the MAC address of PC is found on the MAC address table it forward the message if it's not an ARP request is created to resolve the MAC address for this lab an ARP was created. After the switch forwarded the message to the server and the server and the PC goes into 3 way handshake to established connection-oriented communication from their FTP port 22 on the PC is open. User login with credentials on the server and in every moment both TCP and FTP communication are sent to the server during user authentication from login username to login the password before user is granted permission to either upload or retrieve file.

SKILL GAINED

- Basic network deployment
- IPv4 addressing
- Default gateway configuration cisco IOS CLI usage
- Switch operations
- Connectivity troubleshooting
- Documentation practices

